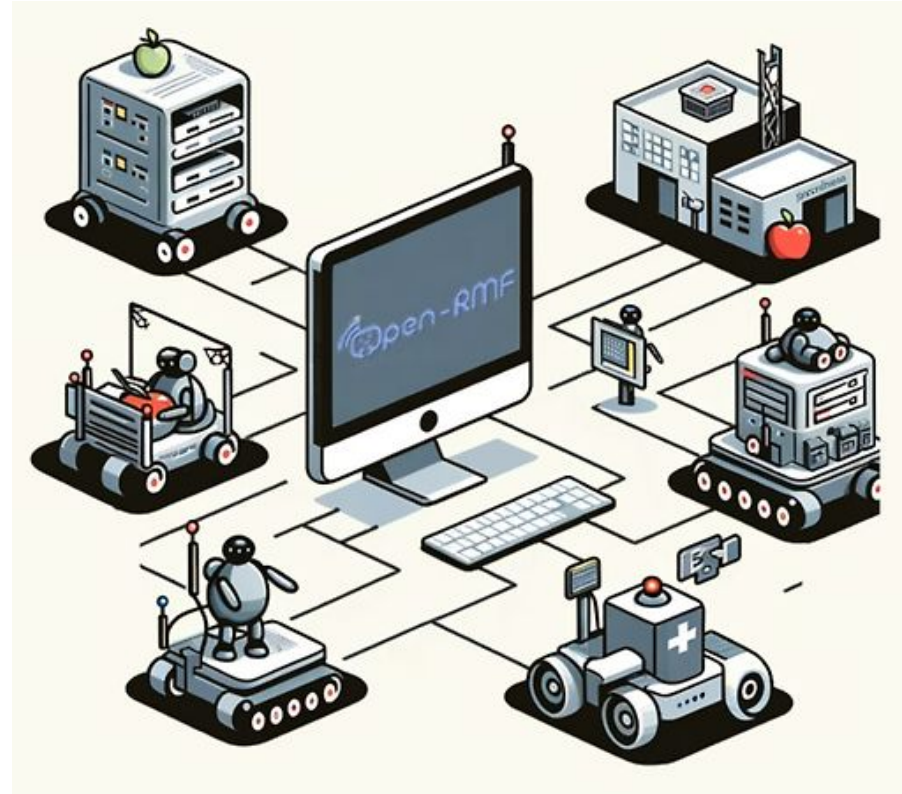


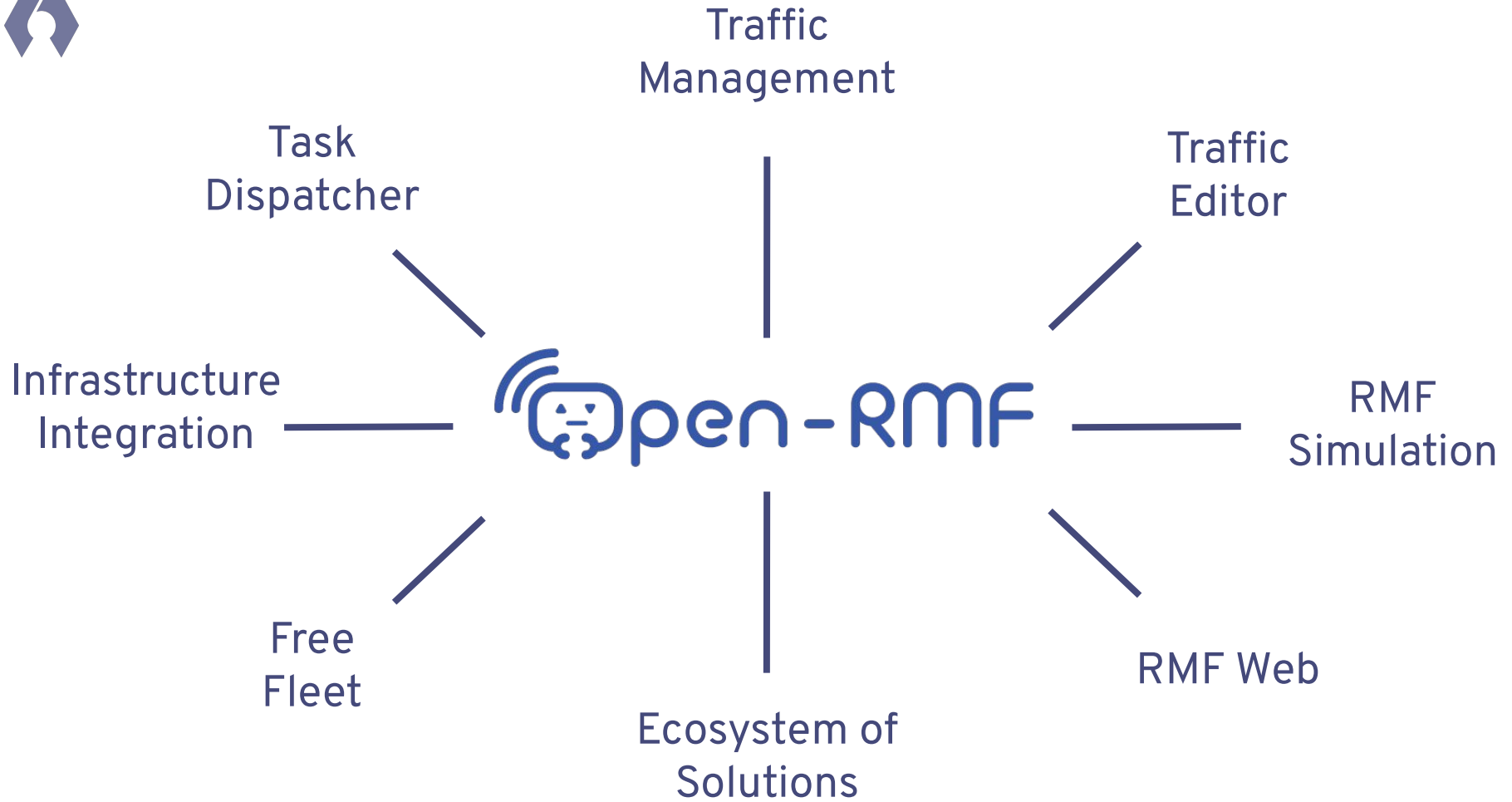
Open-RMF

A Common Language for Robot Interoperability

Open Robotics Middleware Framework (Open-RMF)

The Open Robotics Middleware Framework (Open-RMF) is a free, open source, modular software system that enables sharing and interoperability between multiple fleets of robots and physical infrastructure, like doors, elevators and building management systems.

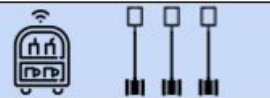
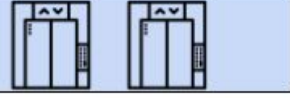


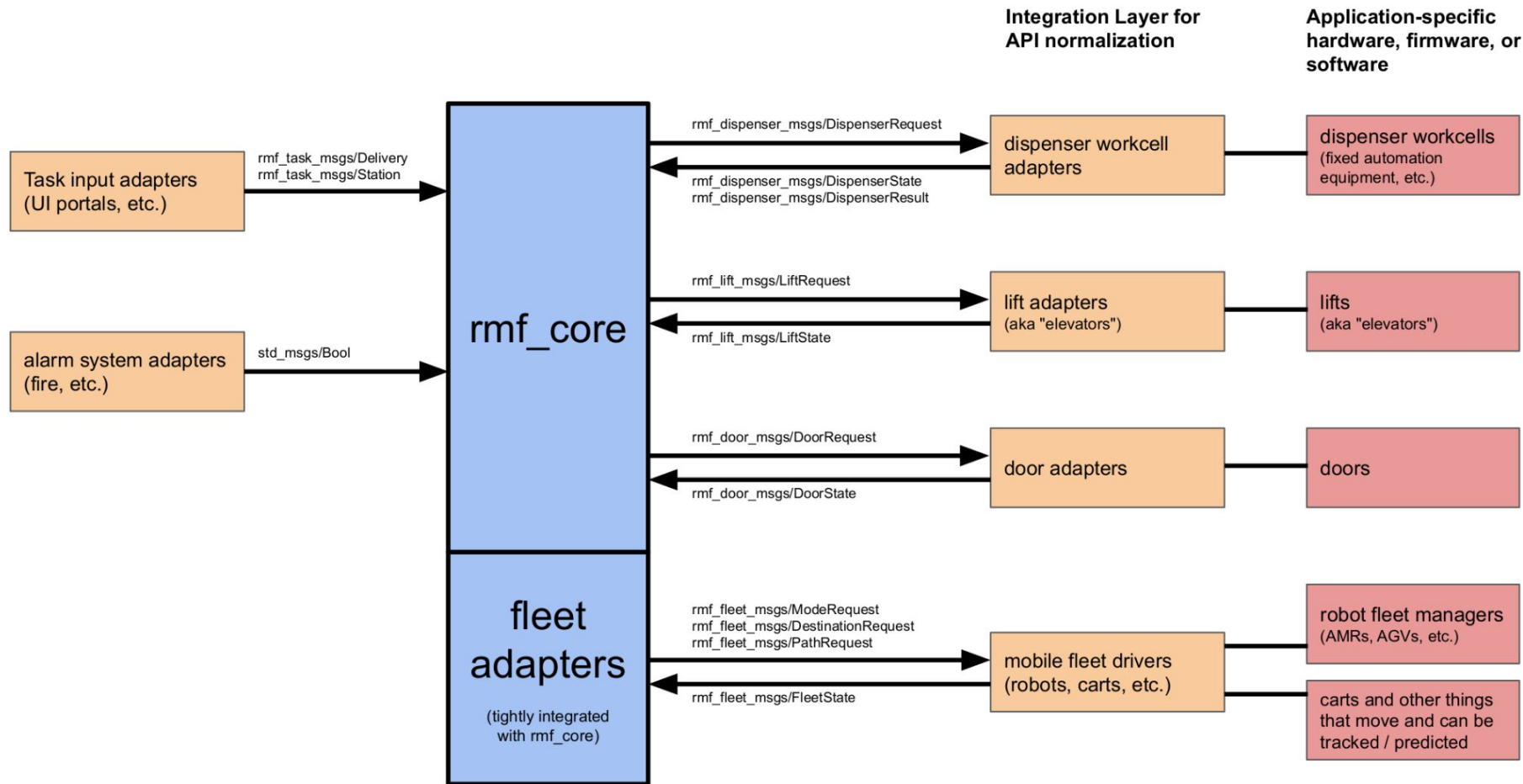


Deployment-specific configuration of Web UI's and bridges to other IT systems

**RMF features
common to all
deployments**

Deployment-specific collection of "RMF adapters" and vendor-provided "managers"

 Web UI Config 1		 Unified Robot Schedule	Fleet Adapter 1	Fleet Manager 1	
 Web UI Config 2			Fleet Adapter 2	Fleet Manager 2	
⋮			⋮	⋮	
 Web UI Config N			Fleet Adapter N	Fleet Manager N or FreeFleet	
 Other System 1	Bridge 1	 Robot Traffic Negotiation	Device Adapter 1	Device Manager 1	
 Other System 2	Bridge 2		Device Adapter 2	Device Manager 2	
⋮	⋮		⋮	⋮	
 Other System N	Bridge N		Device Adapter N	Device Manager N	





Retail



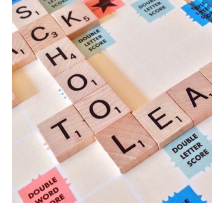
Construction



Healthcare



Airports



Education



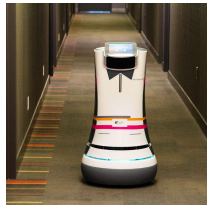
Seaports



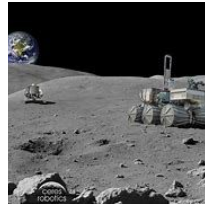
Manufacturing & Logistics



Smart Urban



Hospitality



Space



Agriculture



Office

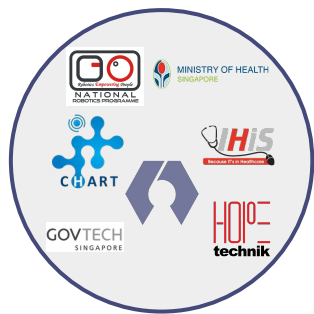
Open-RMF

Open-RMF was initially developed for the **healthcare industry**, but it's equally relevant and useful for **offices, airports, seaports, shopping malls, factories, distribution centers, and more**. There are demos of Hotel, Office, Airport and Clinic environments available to try out!

[See the demos](#)

[Read the Wired article](#)

An (Abridged) History



RMF
collaboration
and research
grant launched
in Singapore

2018

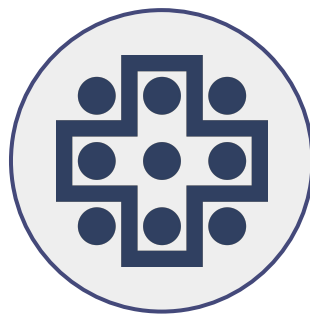


RoMi-H
officially
launched

First demos
of RMF
released

ROSCon
Macau!

2019



RMF 1.0 released

2020



RBR50 award
and WIRED

Open-RMF
GitHub org
launched

Distributed with
ROS 2 Rolling

RHEL 8 support

2021



ROSCon Workshop!

Open-RMF
deployed in
production sites

2022

Beyond

RMF Utilities

- **RMF Demos**

The demos are demonstrations of the capabilities of RMF in various environments. This repository serves as a starting point for working and integrating with RMF.

- **Free Fleet**

Free Fleet is an open-source robot fleet management system for robot developers who do not have their own fleet manager or who would prefer to use and contribute to an open-source fleet management utility.

- **RMF Web UI**

rmf-web is a configurable web application that provides overall visualization and control over the RoMi-H system. The dashboard is by design more "operator-friendly" compared to the schedule visualizer.

- **Traffic Editor**

Traffic Editor is a GUI for creating and annotating floorplans for use in RMF.

- **RMF Simulation**

rmf_simulation contains the simulation plugins to simulate RMF. Plugins are available in gazebo and ignition.

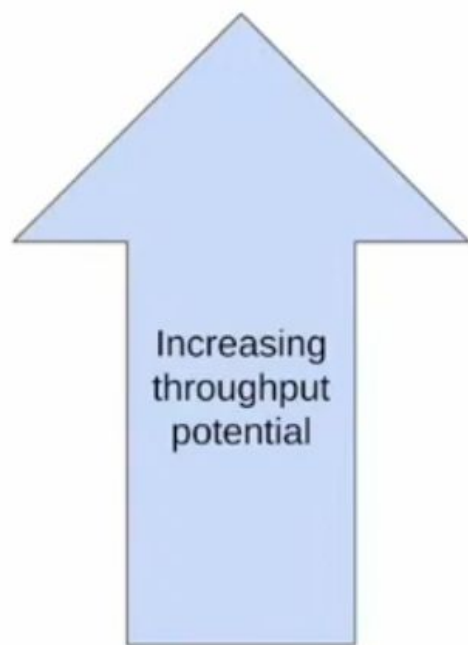
- **Simulation Assets**

The open-source and freely distributable simulation assets are created and shared to accelerate simulation efforts.

- **RMF Schedule Visualizer**

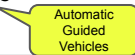
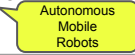
This visualizer is an rviz-based rmf_core visualizer and control panel. It is intended to be a functional tool for RMF developers.

How much control is needed? It's complicated.



Fleet Manager API features	Potential benefits
"paths": robot waypoint control	Reduce stoppages, increase interaction efficiency. Deal gracefully with the unexpected.
"traffic light": robot pause, resume	Spatio-temporal separation of robots can often be achieved.
"read-only": robot locations, destinations	Unified dashboard for operators. Data can allow other robots to avoid conflicts with this fleet.
no fleet API ☹️	None. Fleets cannot coordinate with each other at all to avoid or resolve conflicts.

Interoperability Standards

	Mass Robotics InterOp	VDA 5050	Open-RMF
Provides	Open Specification and reference implementation	Open Specification	FOSS libraries, modules, and an SDK to assist with system integration across multiple robot fleets + building infrastructure
Architecture	Distributed	Centralized Command & Control	Distributed, Negotiation-Driven, Extensible
AGVs 	Position and basic robot state updates	Commanded to move through a shared, predetermined grid	Three choices of integration: (1) Full Control, (2) Traffic Light, and (3) Read-Only
AMRs 	Same as AGVs	Excluded from the standard	Same integration choices as AGVs, plus an upcoming "native integration" in the works
Manually driven vehicles	Same as AGVs	Excluded from the standard	Read-only integration, allows automated vehicles to give way to manual vehicles
Replanning / Rerouting	Excluded from standard	Can be done at "decision points"	Full Control integration allows for spontaneous replanning + rerouting at any time
Task Requests	Coming in Version 2, auction-style task dispatching	Excluded from the standard	Provides optimal dispatching system for composable, extensible task descriptions. Also supports direct robot task requests.
Door / Elevator Integration	Possibly coming in future version	Excluded from the standard, responsibility of the "master control"	SDK takes care of communication with infrastructure for Full Control integrated systems

DOCUMENTATION

The Book : long form text documentation with information and examples

<https://osrf.github.io/ros2multirobotbook/integration.html>

GitHub Org: the official Open-RMF GitHub organization and it's collected repos including RMF Demos

<https://github.com/open-rmf>

Overview: website providing an overview of Open-RMF and some useful links

<https://www.open-rmf.org/>





Urban



Construction



Healthcare



Energy



Education



Transport

Open-RMF



Manufacturing &
Logistics



Smart Cities



Robotics



Space



Agriculture



Cyber



ROS Industrial
Conference 2022

